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Editorial

The need for the MDR (Medical Device Regulation)

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Editorial

It is obvious that the new regulation, Regulation (EU) 2017/745 named Medical Device Regulation or simply MDR (1) of the European Commission, represents a serious problem for many companies manufacturing medical devices that will have to adapt their productive systems, and especially their safety control, monitoring follow-up and notification of possible adverse effects in a very different manner.

The MDR is going to involve a series of very important changes, with the ultimate goal of ensuring that when a product with the CE marking is used, patients and doctors can have sufficient guarantees that the product is safe.

I do not know what has happened in other fields of medicine, but in the specific aspect of medical devices used in intraocular surgery, the behavior of many companies in recent years has been a complete scandal, producing hundreds, probably thousands of blind people worldwide and creating a terrible feeling of insecurity into the eye surgeons.

I insist that the problem is not exclusive to Ophthalmology and a few years ago, the International Consortium of Journalists published a very interesting report (2) collecting cases of problems with medical devices in any field of Medicine, some of them with terrible consequences.

Now the situation has become much more complicated for manufacturers. There are new rules for classifying their products. A new coding system called UDI should be applied to each one. Companies are going to carry out preclinical and clinical studies according to new regulations. And, these studies should be directed by clinicians with demonstrated research experience and developed in centers located in the European Union. They will have to do bibliographic searches and meta-analysis of similar products. And many other requirements.

They have to send to the Regulatory Agencies periodic reports, every year or every two years depending on the classification of the product. Therefore, they are obliged to carefully collect the data prospectively and not simply sell the products.

A central European registry named Eudamed (https://ec.europa.eu/tools/eudamed/#/screen/home) has been created where they must report any undesirable or unforeseen effects and the corrective measures they have taken.

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And they should provide trend reports to the Regulatory Agencies. In other words, they must monitor the problems derived from their products and check how much they deviate from other similar ones on the market. Last but not least, they have to provide more and more accurate information about each product to users both in the information sheets of each container and through the web pages of the companies.

All this implies a huge workload, and the need for companies to collaborate much more with clinical researchers and therefore opens a window of opportunity.

The European Union can never again afford scandals and severe complications with permanent vision loss such as those produced by liquid perfluorocarbons used in retinal detachment surgery, to give just one example (3).

The control over companies is going to be much stricter, but like everything in life, this can have a negative aspect and that is that the appearance of new innovative products that replace the old ones could be slowed down, improving their performance and reducing their complications.

Once again, I am going to turn to a simple example from my specialty. Silicone oil has been used as a long-lasting vitreous substitute since the 1960s. (4) and thousands of patients maintain their vision thanks to this product developed by Paul Cibis in Saint Louis in 1951.

It has solved many problems, but it is the source of numerous complications because among its properties it is a lipophilic compound and therefore capable of altering the cell membranes inside the eye. (5)

The chemical industry has made a considerable effort to purify it, and to reduce among other things the so-called low molecular weight components (LWMC). But they have forgotten that it is an "oil" and therefore has a tendency to emulsify. (6)

Instead of accepting the reality and investing part of their profits in looking for different chemical compounds that are not lipophilic in nature, some companies have engaged in a battle for control of the market, arguing that their products are safe because they are the purest.

Or suggesting that we, ophthalmologists, do accelerate chemistry courses to better select what we are going to use in the patients. (7) As if we weren't challenged enough to correctly diagnose and treat as effectively as possible to our patients.

And when an international director of one of these companies has been asked about cases of toxicity and blindness caused by silicone oil, which all retinologists know, comment on, and present at conferences and in publications, the answer has been: “we have no documented cases”. We say in Spanish that "there is no worse deaf person than the one who does not want to hear." Let's see if now with the implantation of the MDR, which is going at a good pace, they begin to listen to the clinicians as well as to the chemists and they learn the difference between chemical purity and biological safety.

Keywords

Medical Device Regulation

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